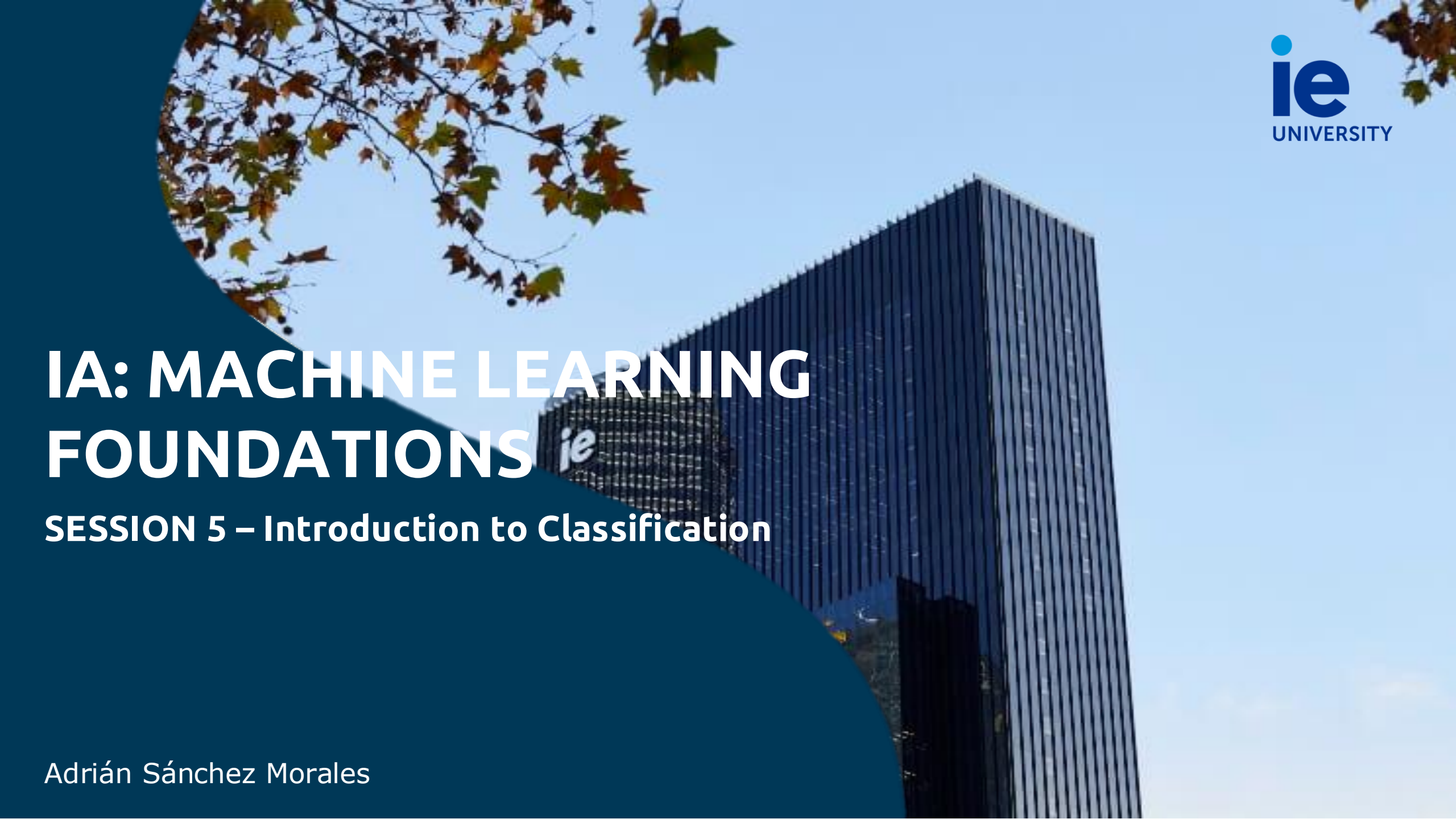


IA: MACHINE LEARNING FOUNDATIONS

SESSION 5 – Introduction to Classification

Adrián Sánchez Morales

AGENDA

1. Why not Linear Regression?

2. Logistic Regression
   2.1. The Logistic Model
   2.2. Multiple Logistic Regression

3. Generative Models
   3.1. Linear Discriminant Analysis
   3.2. Quadratic Discriminant Analysis

4. K-Nearest Neighbors

5. Comparison of Classification Models



ie UNIVERSITY Driving Innovation

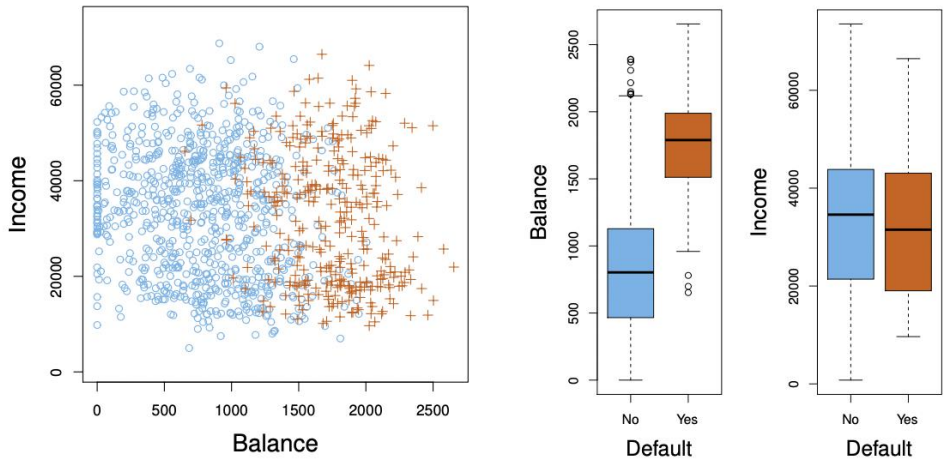
1. WHY NOT LINEAR REGRESSION

- Classification problems are more frequent than regression ones
- In fact, it is very common to transform a regression problem into classification for simplicity (IoT, Fraud Detection, etc.)
- There are at least two reasons not to perform classification using a regression method:

a) A regression method cannot accommodate a qualitative response with more than two classes
b) A regression method will not provide meaningful probabilities of classification

Default Dataset

Whether an individual will default on his or her credit card payment



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2. LOGISTIC REGRESSION

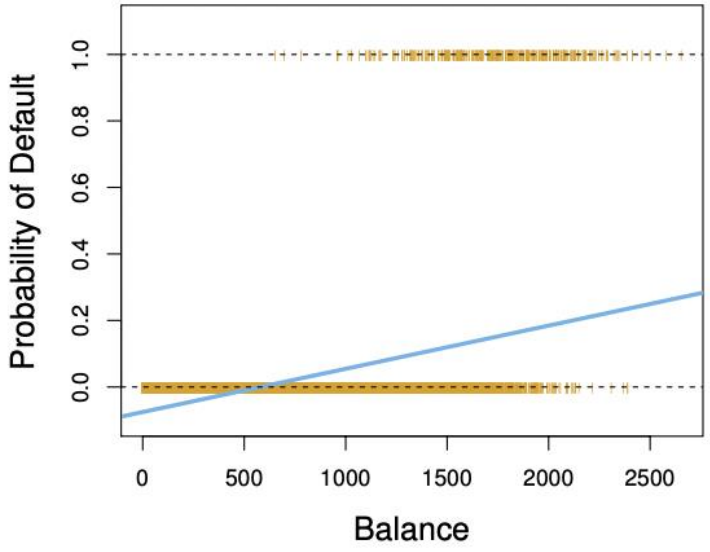
2.1. THE LOGISTIC MODEL

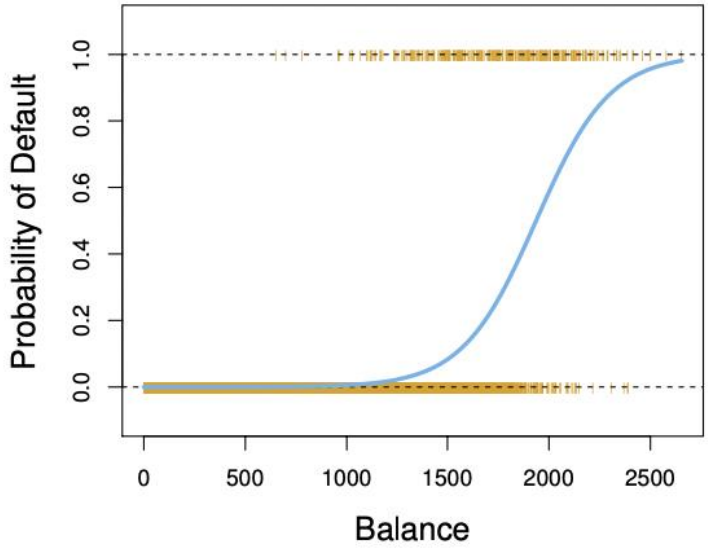
- The Logistic Regression models the *probability* that *Y* belongs to a particular category.

Default Dataset → $\Pr(\texttt{default} = \texttt{Yes}|\texttt{balance})$ or $p(\texttt{balance})$

Regression Model

$$p(X) = \beta_0 + \beta_1 X$$





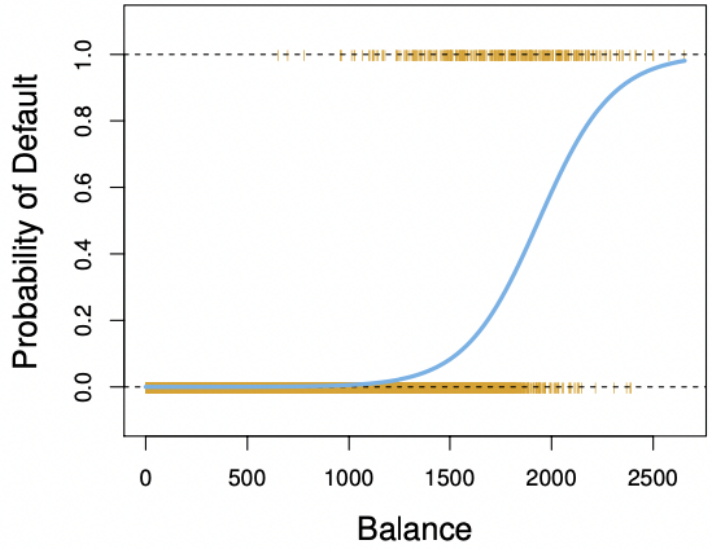
Logistic Function

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

2. LOGISTIC REGRESSION

2.1. THE LOGISTIC MODEL

- The function will always produce *S-shaped* curves and sensible predictions regardless the value of *X*.
- It better captures the range of probabilities
- How do we estimate the regression coefficients?

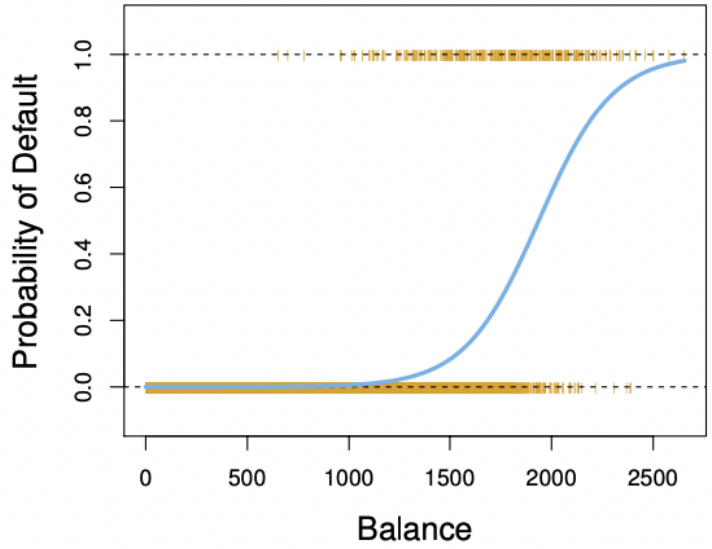


2. LOGISTIC REGRESSION

2.1. THE LOGISTIC MODEL

- The function will always produce *S-shaped* curves and sensible predictions regardless the value of *X*.
- It better captures the range of probabilities
- How do we estimate the regression coefficients?

Maximum Likelihood



- We try to find coefficients such that we get a probability close to one for those who defaulted, and close to zero for those who did not.

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'})).$$

Maximize the Likelihood Function → Software

2. LOGISTIC REGRESSION

2.1. THE LOGISTIC MODEL

- Once the coefficients have been estimated, we can compute the probability for an input using the coefficients:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}}$$

EXERCISE **Default Dataset**

	Coefficient
Intercept	?????
balance	

2. LOGISTIC REGRESSION

2.1. THE LOGISTIC MODEL

- Once the coefficients have been estimated, we can compute the probability for an input using the coefficients:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}}$$

EXERCISE **Default Dataset**

	Coefficient
`Intercept`	−10.6513
`balance`	0.0055

2. LOGISTIC REGRESSION

2.2. MULTIPLE LOGISTIC REGRESSION

- We now consider the problem of predicting a binary response using multiple predictors

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

→ **Maximum Likelihood**

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062

2. LOGISTIC REGRESSION



MULTINOMIAL LOGISTIC REGRESSION

EXERCISE

Use Scikit Learn to:

a) Create a random classification dataset with 1000 rows, 10 input variables and 3 classes
b) Define a multinomial logistic regression model
c) Evaluate the model. Run the example 100 times and compare the average outcome

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3. GENERATIVE MODELS

3.1. LINEAR DISCRIMINANT ANALYSIS

- Why do we need other methods if we have Logistic regression?
  - When there is substantial separation between the two classes, the parameter estimates for the logistic regression model are surprisingly unstable.
  - The methods in this section can be naturally extended to the case of more than two response classes.
  - If the distribution of the predictors X is approximately normal in each of the classes and the sample size is small, then the approaches in this section may be more accurate than logistic regression.

- Everything starts from the Bayes Theorem*

- We make some assumptions:
  - We only have one predictor
  - The PDF of the classes is *Gaussian*:
  - There is a shared variance term across all K classes: $\sigma_1^2 = \cdots = \sigma_K^2 = \sigma^2$

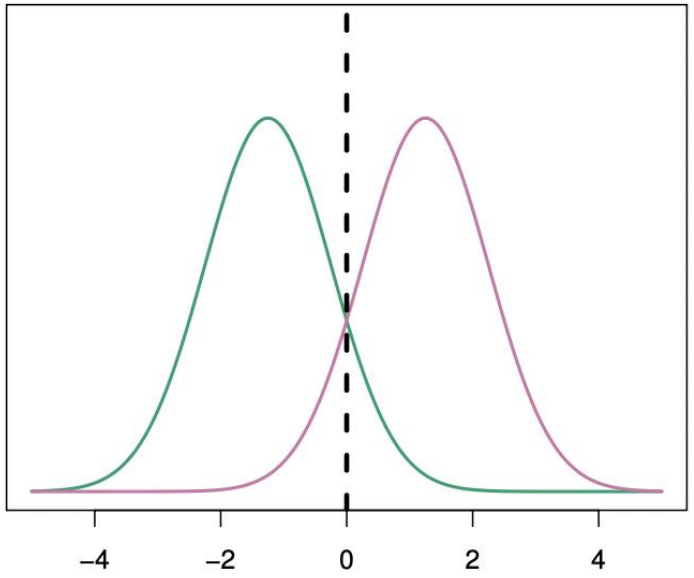
$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)$$

*NOTE: Find more details in the book ITSL

3. GENERATIVE MODELS

3.1. LINEAR DISCRIMINANT ANALYSIS

Bayes Classifier



$$\mu_1 = -1.25$$

$$\mu_2 = 1.25$$

$$\sigma_1^2 = \sigma_2^2 = 1$$

- In practice, we need to estimate:

$$\mu_1, \ldots, \mu_K \qquad \sigma^2$$

*NOTE: Find more details in the book ITSL

3. GENERATIVE MODELS

3.1. LINEAR DISCRIMINANT ANALYSIS

- The linear discriminant analysis (LDA) method approximates the Bayes classifier by:

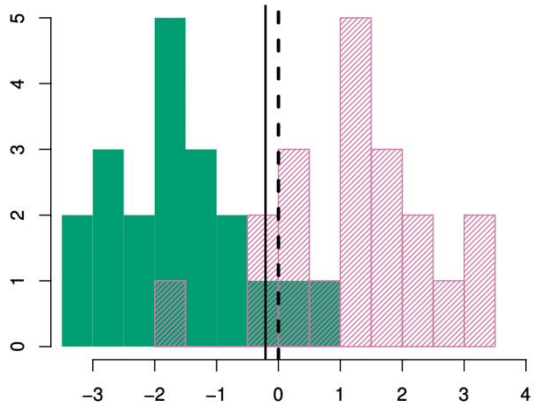
$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

→ Simply the average of all the training observations from the k_{th} class

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

→ A weighted average of the sample variances for each of the *K* classes

- Imagine that we have 20 observations from the previous classes

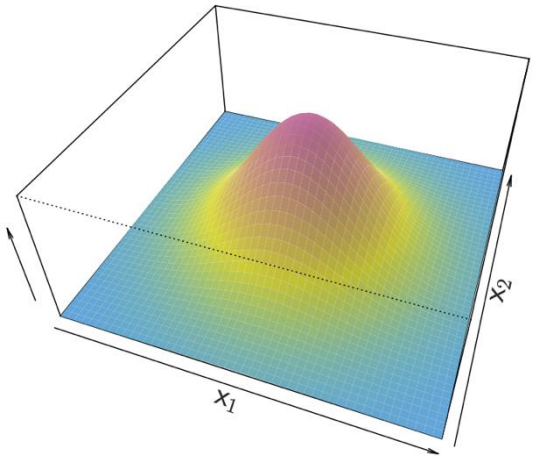


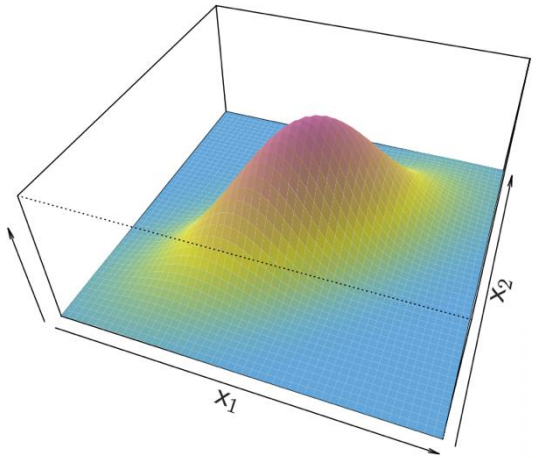
1. Compute $\hat{\mu}_k$
2. Compute $\hat{\sigma}^2$
3. We determine the class by using:

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k) \qquad \hat{\pi}_k = n_k/n$$

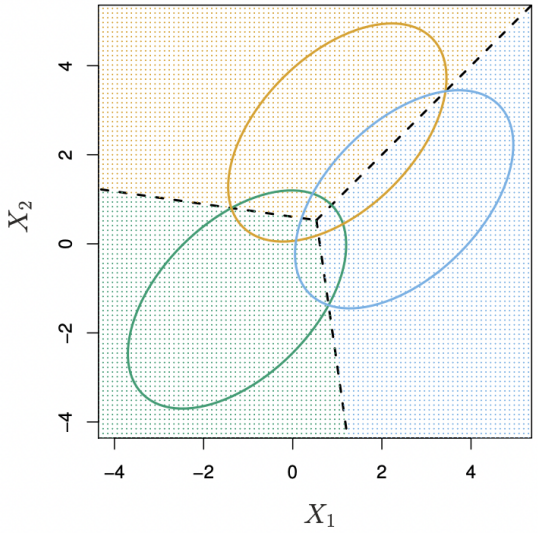
3. GENERATIVE MODELS

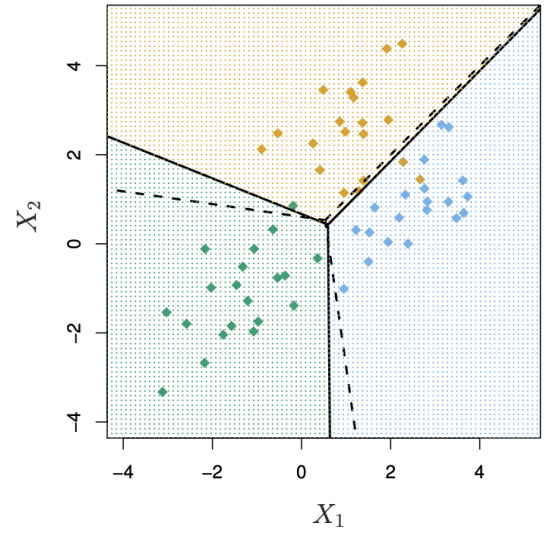
3.1. LINEAR DISCRIMINANT ANALYSIS





$$X \sim N(\mu, \Sigma)$$





3. GENERATIVE MODELS

3.2. QUADRATIC DISCRIMINANT ANALYSIS

- Like LDA, the QDA classifier results from assuming that the observations from each class are drawn from a Gaussian distribution. However, **QDA assumes that each class has its own covariance matrix**.

$$\begin{aligned}\delta_k(x) &= -\frac{1}{2}(x-\mu_k)^T\mathbf{\Sigma}_k^{-1}(x-\mu_k) - \frac{1}{2}\log|\mathbf{\Sigma}_k| + \log\pi_k \\ &= -\frac{1}{2}x^T\mathbf{\Sigma}_k^{-1}x + x^T\mathbf{\Sigma}_k^{-1}\mu_k - \frac{1}{2}\mu_k^T\mathbf{\Sigma}_k^{-1}\mu_k - \frac{1}{2}\log|\mathbf{\Sigma}_k| + \log\pi_k\end{aligned}$$

- Unlike LDA, now the quantity *x* appears as a *quadratic* function.

3. GENERATIVE MODELS

3.2. QUADRATIC DISCRIMINANT ANALYSIS

DISCUSSION

Why would one prefer LDA to QDA, or viceversa?

3. GENERATIVE MODELS

3.2. QUADRATIC DISCRIMINANT ANALYSIS

DISCUSSION

Why would one prefer LDA to QDA, or viceversa?

The answer lies in the **bias-variance trade-off**. When there are *p* predictors, then estimating a covariance matrix requires estimating *p(p+1)/2* parameters. QDA estimates a separate covariance matrix for each class, for a total of *Kp(p+1)/2* parameters. With 50 predictors this is some multiple of 1.275, which is a lot of parameters. By instead assuming that the *K* classes share a common covariance matrix, the LDA model becomes linear in *x*, which means there are *Kp* linear coefficients to estimate. Consequently, LDA is a much less flexible classifier than QDA, and so has substantially lower variance. This can potentially lead to improved prediction performance. But there is a trade-off: if LDA's assumption that the K classes share a common covariance matrix is badly off, then LDA can suffer from high bias. Roughly speaking, **LDA tends to be a better bet than QDA if there are relatively few training observations and so reducing variance is crucial. In contrast, QDA is recommended if the training set is very large**, so that the variance of the classifier is not a major concern, **or if the assumption of a common covariance matrix for the K classes is clearly untenable**.

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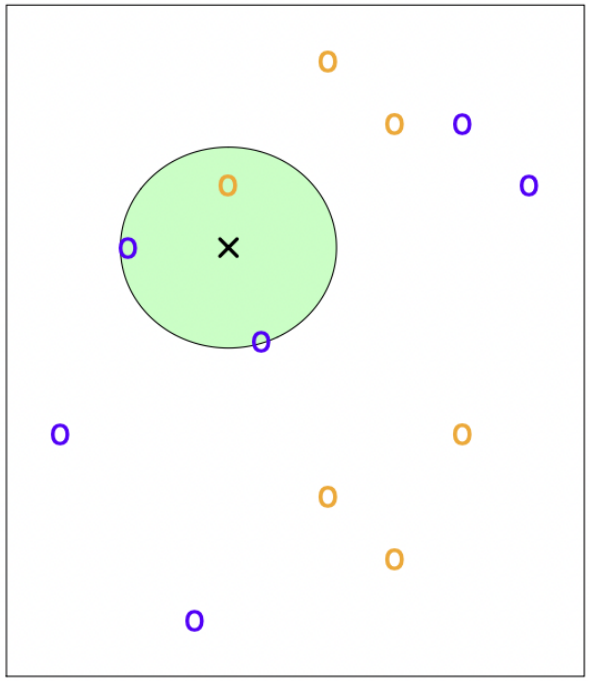
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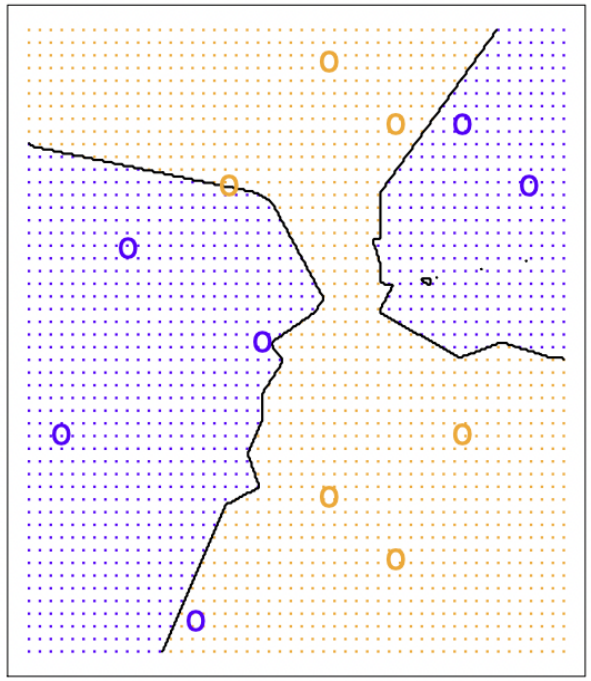
5. Comparison of Classification Models



4. K-NEAREST NEIGHBORS

- Given a positive integer *K* and a test observation x_0, the KNN classifier first identifies the *K* points in the training data that are closest to x_0.

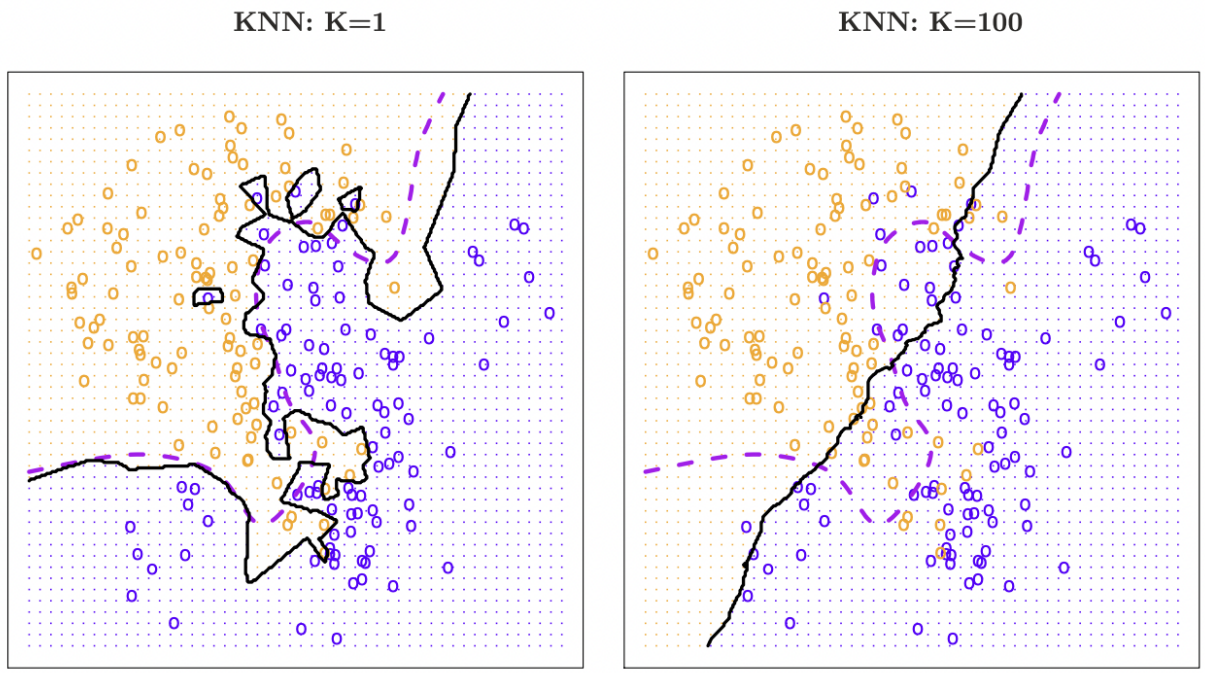




$$\Pr(Y = j|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

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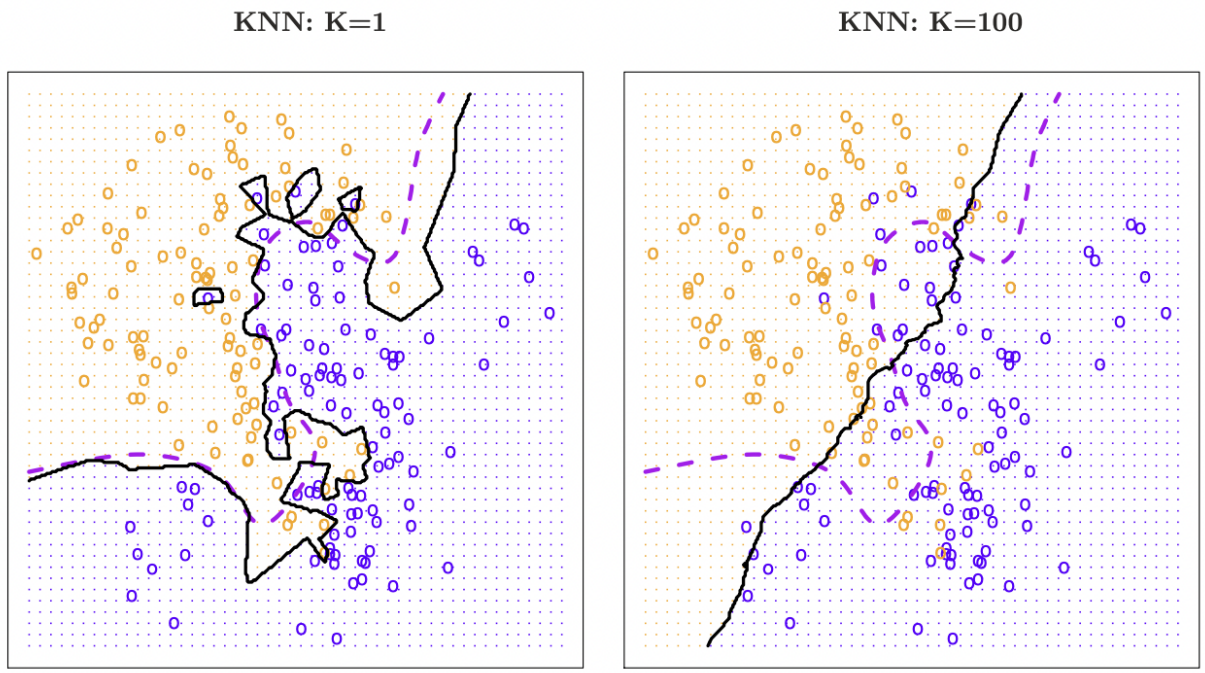


- With *K = 1*, the decision boundary is overly flexible
- With *K = 100,* it is not sufficiently flexible.

Optimal Value of K?

4. K-NEAREST NEIGHBORS

- Given a positive integer *K* and a test observation x_0, the KNN classifier first identifies the *K* points in the training data that are closest to x_0.



- With *K = 1*, the decision boundary is overly flexible
- With *K = 100,* it is not sufficiently flexible.

Optimal Value of K?

- Data scientists usually choose as an odd number for binary problems and another simple approach is:

$$\boldsymbol{k = \sqrt{n}}$$

- Cross Validation

4. K-NEAREST NEIGHBORS

- *Simple and intuitive.* It is simple and easy to explain to laypeople.
- *Non-parametric*, therefore, it doesn't have any assumptions on the data distribution
- *No training step*: KNN is more of an exception to the general machine learning workflow. It doesn't have a training/validation/test set. The model created with KNN is available in a labeled dataset, placed in metric space. Say, if you want to classify any object, the model has to read through all the data and compare the distances of the closest objects.
- *Easy to implement for multi-class problems:* Compared to other algorithms, it is very easy to predict multiclass problems. Just supply the 'k' a value that is equivalent to the number of classes and you are good to go.
- *Few hyperparameters:* When working with K-NN, you just need to provide two parameters, k (the numbers of neighbors to consider) and the choice of Distance Function (e.g. Euclidean, Manhattan distance).
- *Used for classification and regression:* It can be used for classification and regression
- *Instance-based learning (lazy learning):* You don't need to fit a model in advance, just provide the data point and it will give you the prediction.

4. K-NEAREST NEIGHBORS

▼ Slow with a larger dataset. If it is going to classify a new sample, it will have to read the whole dataset, hence, it becomes very slow as the dataset increases.

▼ *Curse of dimensionality:* KNN is more appropriate to use when you have a small number of inputs. If the number of variables grows, the KNN algorithm will have a hard time predicting the output of a new data point.

▼ *Feature inputs need to be scaled:* It is a *must* that the features should be scaled. KNN uses distance criteria, like Euclidean or Manhattan distances, therefore, it is very important that all the features have the same scale.

▼ *Outlier sensitivity:* KNN is very sensitive to outliers. Since it is an instance-based algorithm based on the distance criteria, if we have some outliers in the data, it is liable to create a biased outcome.

▼ *Missing Value not treated:* It is not capable of treating or dealing with missing values

▼ *Class imbalance can be an issue:* If we have an imbalanced class data, the algorithm might wrongly pick the majority class.

4. K-NEAREST NEIGHBORS

PRACTICE

Take a look and understand the following tutorial:

"Develop k-Nearest Neighbors in Python From Scratch"

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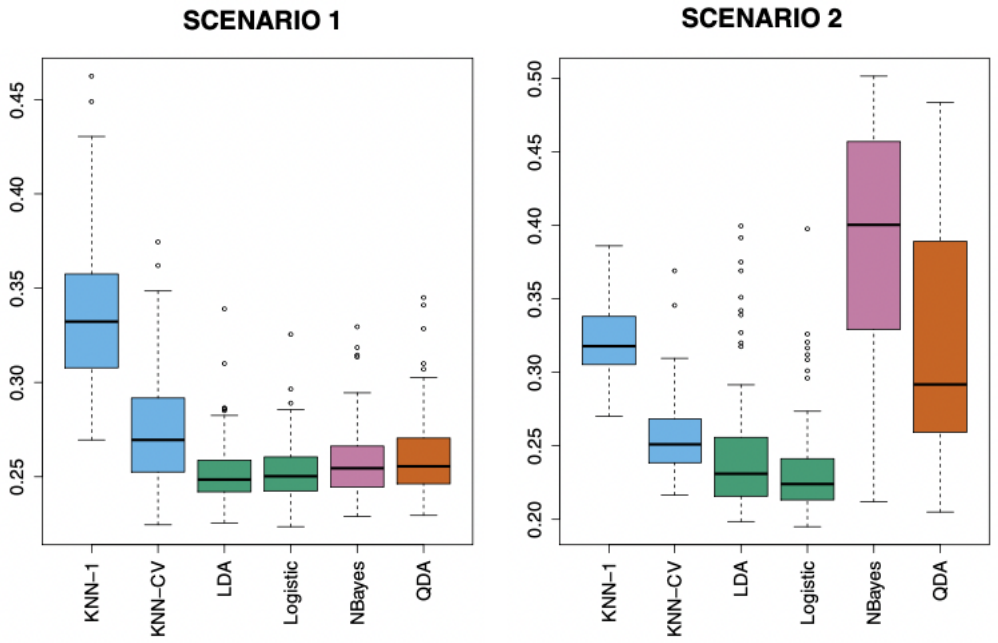
5. Comparison of Classification Models



5. COMPARISON OF CLASSIFICATION MODELS

- For a mathematical comparison go to the recommended books of the subject.
- Let's see a practical comparison with the **test error representation**: binary classification, 100 random training samples, 2 quantitative predictors

- Classes from uncorrelated random normal variables with a different mean in each class



- Negative correlation between predictors
- Outliers

*NOTE: Find more details in the book ITSL


THANK YOU